Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 32. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the training loss.

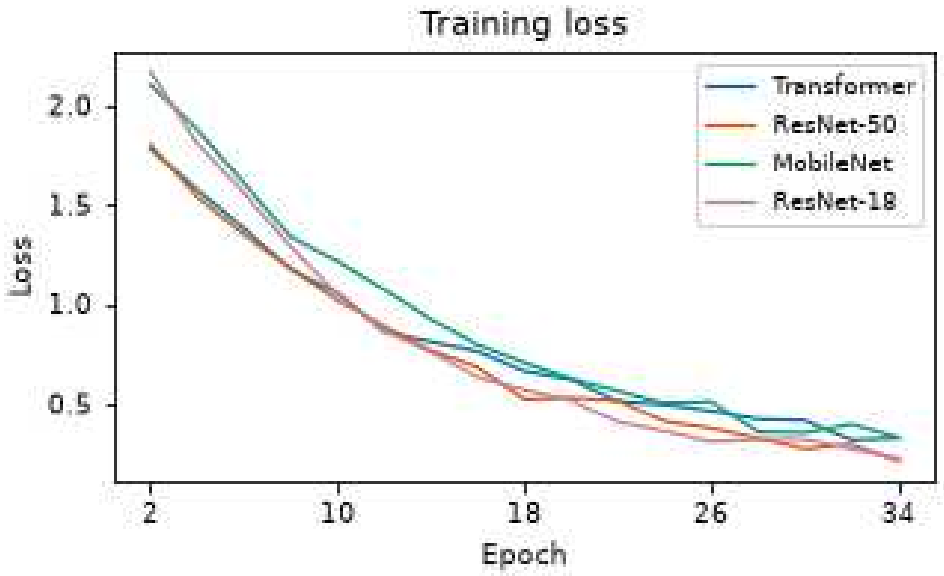


Figure 1: Training loss for the models in this experiment.

At epoch 22, the loss of ResNet-18 is 0.622. MobileNet has a loss of 0.394 at epoch 32. ResNet-50 has a loss of 0.101 at epoch 34 (Figure 1).

Figure 2 shows the training loss.



Figure 2: Training loss for the models in this experiment.

MobileNet has a loss of 1.031 at epoch 2. At epoch 3, the loss of GRU is 0.956 (Figure 2). At epoch 4, the loss of Ours is 0.283. MobileNet has a loss of 0.956 at epoch 2 (Figure 2).

Figure 3 shows the training loss.

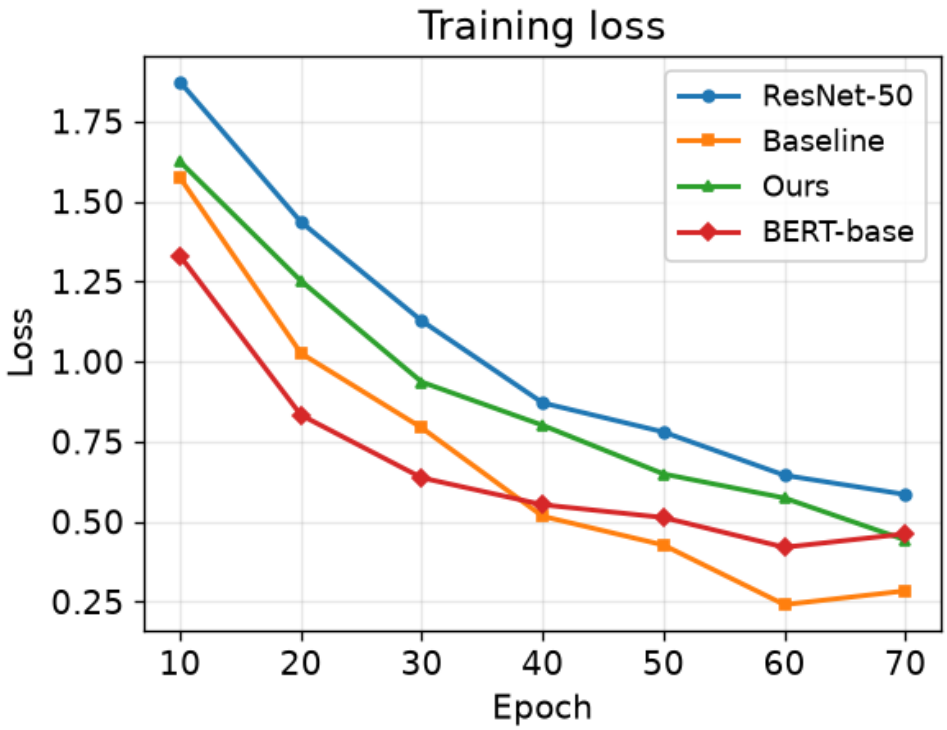


Figure 3: Training loss for the models in this experiment.

Ours has a loss of 0.936 at epoch 30. At epoch 40, the loss of BERT-base is 0.733.

3 Discussion

On IMDB, Naive Bayes obtains an accuracy of 87.8%. The differences between the models are discussed in the next section.